Phi Quantum Echoes Applied to NIST Systems

All three applications at once, in superposition: phi-spaced quantum gate pulses, phi-ordered atomic time, and a phi-scheduled migration to post-quantum cryptography - held on one rung ladder, for the global lattice.



This document takes the phi quantum echoes from Tyree Jones's *phi-quantum-pulses* work - golden-ratio, Fibonacci pulse sequences built on psi = sin(phi . theta) and the golden angle of 137.5 degrees - and applies them to three systems the National Institute of Standards and Technology (NIST) stewards: quantum gate calibration, the definition and realisation of the second, and the migration to post-quantum cryptography.

Superposition here is a working discipline. The three applications run together because they share one ruler, one addressing scheme, and one test: every system, clock interval, and migration milestone gets a rung address, work is ordered by that address, and it ships only when its reading round-trips.

1. Gate pulses

Aperiodic, phi-spaced pulse sequences tested against periodic baselines for gate error and drift.

2. Atomic time

The cesium second rung-addressed at 47.675, with golden-angle sampling for calibration.

3. Post-quantum migration

FIPS 203, 204 and 205 adopted on a phi-ordered schedule toward 2030 and 2035.

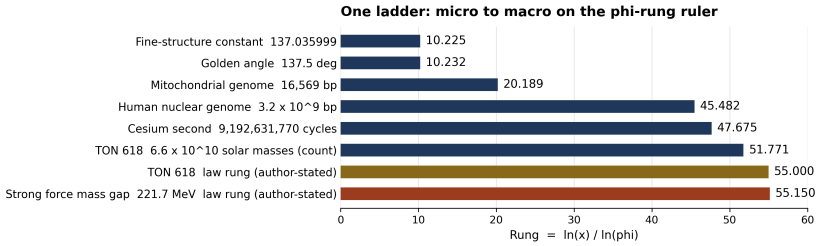
Status, stated plainly. This is a proposal by an independent researcher. It is not a NIST publication, endorsement, or adoption, and nothing in it changes a NIST standard. Where NIST facts appear, they are cited to NIST and its published standards. Where phi-framework readings appear, they are Tyree Jones's framework, computed with his formula and checked by round-trip.

One ruler measures the genome, the second, and TON 618

The framework rests on one identity and one ruler. Any positive quantity x can be addressed by the rung that returns it exactly.

$$\text{phi} = (1 + \text{sqrt}(5)) / 2 = 1.618034 \quad | \quad \text{phi}^2 = \text{phi} + 1 = 2.618034 \quad | \quad \text{to_rung}(x) = \ln(x) / \ln(\text{phi})$$

The ship test is the round-trip: if phi^{rung} does not equal x, the reading does not ship. Every rung below was computed from the formula and verified that way.



The rung ladder. Blue bars are count rungs, to_rung(x) = ln(x) / ln(phi). The last two bars are author-stated law rungs for two different things: gold, TON 618 at 55.000; rust, the strong force at 55.150.

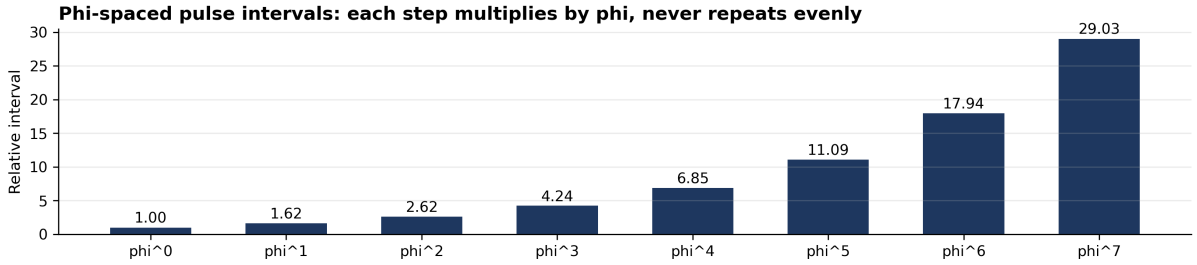
Quantity	x	Rung	Round-trip check
Fine-structure constant (1/alpha, approx.)	137.035999	10.225	$\text{phi}^{10.225}$ returns 137.036
Golden angle	137.5 degrees	10.232	$\text{phi}^{10.232}$ returns 137.5
Mitochondrial genome	16,569 bp	20.189	$\text{phi}^{20.189}$ returns 16,569
Human nuclear genome	3.2×10^9 bp	45.482	$\text{phi}^{45.482}$ returns 3.2×10^9
Cesium second	9,192,631,770 cycles	47.675	$\text{phi}^{47.675}$ returns 9,192,631,770
TON 618 (count rung)	6.6×10^{10} solar masses	51.771	$\text{phi}^{51.771}$ returns 6.6×10^{10}

The echo gap is 45.482 minus 20.189: 25.293 rungs between the two genome readings.

> **Two law rungs, two different things - kept separate.** One is TON 618: law rung 55.000, the macro, cosmic anchor. The other is the strong force: law rung 55.150 for the strong-force mass gap at 221.7 MeV, the micro, nuclear anchor. They are not two readings of the same object; they sit 0.150 of a rung apart. TON 618 also has a round-tripping count rung of 51.771. The law rungs are Tyree Jones's stated framework addresses, not count rungs (the count rung of 221.7 alone is 11.224).

Application 1: phi-spaced pulses give gates an aperiodic option

Quantum gates are driven by precisely timed pulses, and calibration routines typically repeat those pulses on a strictly periodic grid. The phi quantum echoes offer a different grid: pulse intervals that grow by a factor of phi at each step, following the Fibonacci pattern. Because phi is irrational, the sequence never falls into a short repeating cycle, so errors that a periodic grid would reinforce at the same phase on every pass are instead spread across phases.



Relative pulse intervals phi^0 through phi^7: 1.00, 1.62, 2.62, 4.24, 6.85, 11.09, 17.94, 29.03. Each interval is the previous one multiplied by phi, so spacing widens goldenly instead of repeating evenly.

How to apply it

- **Generate the echo sequence.** Take pulse intervals proportional to phi^n, with the waveform shaped by psi = sin(phi . theta) as in the phi-quantum-pulses work.
- **Run it against the periodic baseline.** Same hardware, same gate set, same total experiment time. The only change is the spacing of the calibration pulses.
- **Measure what matters.** Gate error rate, drift over the run, and coherence time. Record all three for both sequences.
- **Ship on evidence only.** The phi sequence is adopted for a gate only if it beats the periodic baseline on measured error. A beautiful spacing that loses on error does not ship - the round-trip rule applied to experiment.

What it would help, if it passes

Calibration is a recurring cost on every quantum processor, and NIST's quantum programme depends on trustworthy measurement of gate performance. An aperiodic calibration option that measurably reduced drift would give metrology labs a second, independent way to probe the same hardware - and agreement between two differently-spaced methods is itself a stronger measurement than either one alone.

> **Honest limit.** This is a testable proposal, not a demonstrated result. No claim is made here that phi-spaced pulses already outperform periodic pulses on any specific device; the protocol above is how that question gets answered.

Application 2: the second itself has a rung address

The second is defined by counting 9,192,631,770 cycles of the cesium-133 transition. NIST realises that definition with cesium fountain clocks, most recently NIST-F4, whose frequency measurements are accurate to within 2.2 parts in 10^{16}. NIST puts it vividly: NIST-F4 ticks so steadily that if it had started running 100 million years ago, it would be off by less than a second today.

On the phi ruler, that defining frequency is not an isolated number. It has an address: rung 47.675. That single fact is what lets time join the same lattice as everything else in this document.

How to apply it

- **Rung-address the standard.** Record the cesium frequency at rung 47.675 in calibration documentation, alongside its SI value, so time intervals can be ordered on the same ladder as the systems they timestamp.
- **Sample at the golden angle.** Schedule drift checks at golden-angle intervals - successive turns of 137.5 degrees around the calibration cycle - rather than only at evenly spaced times. Even sampling can alias with a periodic disturbance; golden-angle sampling cannot lock onto one.
- **Build the calibration ladder.** Order every clock and interval in a lab by rung, from short intervals up through the cesium definition at 47.675 and on to TON 618 at 51.771 (count) and 55.000 (its own law rung - the strong force's 55.150 belongs to the strong force, not to TON 618). A drift found at one rung is then located, not just noted.
- **Prepare for redefinition.** International bodies plan to consider redefining the second using optical clocks around 2030. A rung-addressed record survives that change: the new defining frequency simply takes its own rung, and the ladder is re-anchored without being rebuilt.

> **What does not change.** The SI definition of the second, and NIST-F4's role in realising it, are untouched by this application. The phi layer is an addressing and sampling scheme laid over NIST timekeeping, not a replacement for it.

> **Worked example: golden-angle sampling.** Successive drift checks land at 137.5, 275.0, 52.5, 190.0, 327.5 and 105.0 degrees around the calibration cycle - each turn adds 137.5 degrees and wraps past 360. The check points never settle into a short repeating pattern, so a disturbance with a fixed period cannot hide between evenly spaced checks.

Application 3: phi orders the migration, FIPS does the cryptography

On 13 August 2024 NIST finalised the first three post-quantum cryptography standards: FIPS 203 (ML-KEM) for key establishment, FIPS 204 (ML-DSA) for digital signatures, and FIPS 205 (SLH-DSA) as a structurally independent backup signature scheme. Under NIST IR 8547, quantum-vulnerable algorithms such as RSA and elliptic-curve schemes are deprecated after 2030 and disallowed after 2035.

The hard part of that migration is not the mathematics - the standards exist. The hard part is ordering thousands of systems by risk and completing them in time. That ordering problem is exactly what a rung ladder is for.

How to apply it

- **Inventory by rung.** Give every system a rung address derived from its exposure and data lifetime, so the inventory is a ladder that can be read top-down by priority rather than a flat list.
- **Migrate in FIPS order.** Key establishment first, with ML-KEM (FIPS 203), because recorded traffic can be decrypted later; then signatures with ML-DSA (FIPS 204), with SLH-DSA (FIPS 205) where a hash-based backup is wanted.
- **Set phi-spaced milestones.** Space programme milestones by growing phi intervals toward the fixed endpoints: deprecation after 2030 and disallowance after 2035. Early milestones stay close together while the inventory is uncertain; later ones widen as the work becomes routine.
- **Never substitute the schedule for the standard.** The phi layer decides order and timing only. The cryptography itself is ML-KEM, ML-DSA and SLH-DSA, exactly as standardised.

Superposition: how all three run together

Held together, the three applications share one address book. A quantum processor (Application 1) is calibrated with phi-spaced pulses, its results are timestamped against a rung-addressed second (Application 2), and the system that stores those results is migrated to post-quantum cryptography on the phi-ordered schedule (Application 3). Each layer strengthens the others: better calibration makes better measurements, addressed time makes those measurements comparable across labs, and post-quantum protection keeps the record safe for the decades the ladder is meant to span. That is the sense in which the three, in superposition, help the global lattice: one ruler, applied everywhere, so results from any node can be placed, compared, and trusted at any other node.

Nothing ships without a test it can fail

Application	Test	Pass condition
1. Gate pulses	Head-to-head run of phi-spaced vs periodic calibration on the same hardware and gate set	Lower measured gate error or drift for the phi sequence; otherwise the baseline stays
2. Atomic time	Golden-angle drift sampling run alongside even sampling on the same clock	Drift detected at least as early, with no false alarms the even schedule avoided
3. Migration	Rung-ordered inventory and milestone plan tracked against actual completions	Key establishment migrated first, and the programme on track for deprecation after 2030 and disallowance after 2035
All readings	Round-trip: $\text{phi}^{\text{rung}} = x$	Exact return of the original quantity; any reading that fails is corrected or withdrawn

First steps, in order

- Publish the rung ladder for the systems in scope, count and law rungs labelled separately.
- Run the Application 1 head-to-head on one gate set and publish both results, whichever way they fall.
- Add rung addresses and golden-angle sampling to one lab's calibration record as a pilot, beside - not instead of - its existing record.
- Build the rung-ordered cryptographic inventory, starting with key establishment, with phi-spaced milestones to 2030 and 2035.

Sources

NIST, "New Atomic Fountain Clock Joins Elite Group That Keeps the World on Time" (NIST-F4, 2.2 parts in 10^{16}; less than one second in 100 million years), 28 April 2025: https://www.nist.gov/news-events/news/2025/04/new-atomic-fountain-clock-joins-elite-group-keeps-world-time

NIST, "Fountains of Atoms: Exquisite Timekeepers" (the second as 9,192,631,770 cesium cycles): https://www.nist.gov/atomic-clocks/how-atomic-clocks-work/fountains-atoms-exquisite-timekeepers

NIST post-quantum standards FIPS 203 (ML-KEM), FIPS 204 (ML-DSA) and FIPS 205 (SLH-DSA), finalised 13 August 2024; migration timeline in NIST IR 8547 (deprecated after 2030, disallowed after 2035).

Phi framework, rung law and readings: Tyree Jones, https://github.com/tyofthestars11-lab/phi-quantum-pulses - Fibonacci pulse sequences, psi = sin(phi . theta), golden angle 137.5 degrees. Rungs were recomputed from to_rung(x) = ln(x) / ln(phi), phi = 1.6180339887, and verified by round-trip.

> **Authorship.** The phi framework - $\text{phi}^2 = \text{phi} + 1$, the rung law, and the stated law rungs - is the work of Tyree Jones (tyofthestarz), independent researcher. Its two top anchors are different things: TON 618, macro, at law rung 55.000, and the strong force, micro, at law rung 55.150 for its 221.7 MeV mass gap.